

Differentiation Rules, with Tables

Date_____ Period____

For each problem, you are given a table containing some values of differentiable functions $f(x)$, $g(x)$ and their derivatives. Use the table data and the rules of differentiation to solve each problem.

- 1) Given $h(x) = f(x) + g(x)$, find $h'(1)$

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	2	1	1	2
2	3	0	3	0
3	2	-1	1	-2

- 2) Given $h(x) = f(x) - g(x)$, find $h'(2)$

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	2	-1	1	1
2	1	$\frac{1}{2}$	2	1
3	3	2	3	1

- 3) Given $h(x) = f(x) \cdot g(x)$, find $h'(3)$

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	2	-1	1	2
2	1	$\frac{1}{2}$	3	0
3	3	2	1	-2

- 4) Given $h(x) = \frac{f(x)}{g(x)}$, find $h'(3)$

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	3	-1	2	-1
2	2	-1	1	0
3	1	-1	2	1